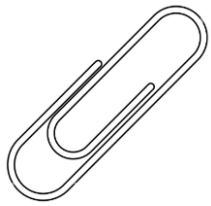
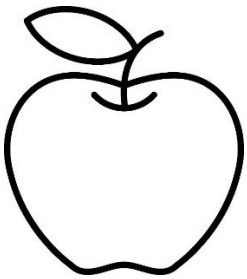
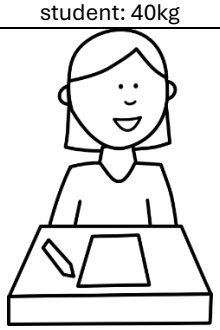
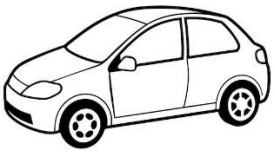
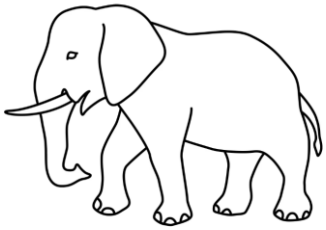
Estimations

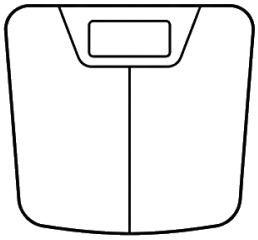
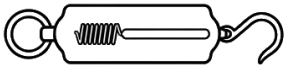
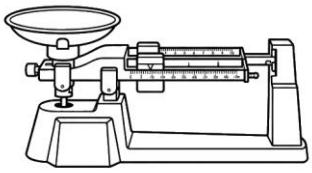
Paper clip: 1g	Large apple: 150g	2L fizzy drink: 2kg	Large bag of potatoes: 7kg
			

Average weight of a grade 6 student: 40kg	Weight of an adult: 80-100kg	Weight of a car: 1000kg	Weight of an elephant: 6000kg
			

1. Estimate the mass of:

1.1)	Mathematics textbook.	8 g	800 g	8 kg	80 kg
1.2)	Single chicken egg.	60 g	600 g	6 kg	60 kg
1.3)	Full-grown domestic cat.	45 g	450 g	4,5 kg	45 kg
1.4)	Single paperclip.	1 g	10 g	100 g	1 kg
1.5)	Slab of chocolate.	1 g	100 g	1 kg	10 kg
1.6)	Standard building brick used to build a house	3 g	30 g	300 g	3 kg
1.7)	Standard bicycle.	12 g	120 g	1,2 kg	12 kg
1.8)	Standard loaf of bread.	7 g	70 g	700 g	7 kg
1.9)	Fully grown male lion.	1,9 kg	19 kg	190 kg	1 900 kg
1.10)	Farm tractor.	270 g	2,7 kg	27 kg	2 700 kg

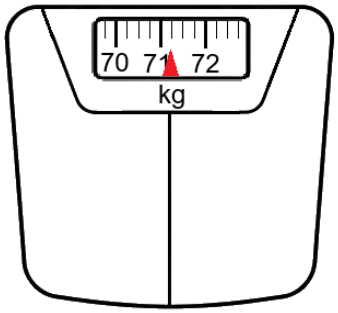
Mass and weight: instruments

Bathroom scale 	**What it does:** Measures heavier objects and people sitting or standing directly on top of it. Use: 1. **Yourself:** Stepping on the scale to check how many kilograms you weigh. 2. **Luggage:** Checking if a packed travel suitcase is under the airline's weight limit before going on a trip.
Balance scale 	**What it does:** It acts like a seesaw; we put the object that we want to weigh on one side and then we put weights on the other side. When both sides **balance**, then we would have found its weight by adding up all the weights used. Use: 1. **Market Fruit:** Balancing a bag of apples against heavy metal weights at a farm stand 2. **Spices or Grains:** Measuring out a scoop of rice or coffee beans at a market.
Kitchen scale 	**What it does:** Measures light to medium-weight ingredients and everyday food items, usually in grams or small kilograms. Use: 1. **Baking:** Measuring 250 g of flour or butter so a cake recipe turns out perfectly. 2. **Small Liquids/Packages:** Checking the weight of a bottle of juice or a small parcel you want to post.
Spring scale 	**What it does:** Uses a stretchy spring and a hook underneath to measure weight or the force of pulling an object. Use: 1. **Fishing:** Hanging a freshly caught fish from the hook to see how heavy it is. 2. **Force:** demonstrates the force needed to pull an object
Triple Beam Balance 	**What it does:** Uses three sliding metal weights along labelled bars (beams) to find very precise measurements down to tiny fractions of a gram. **Use:** Finding the exact mass of a small rock, coin, or metal key in a science lab.

Questions

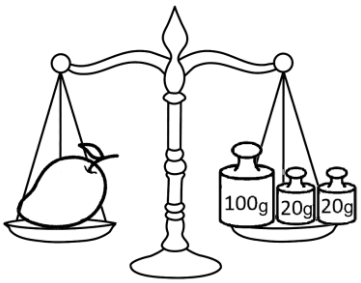
2. What are the readings of the instruments?

a)



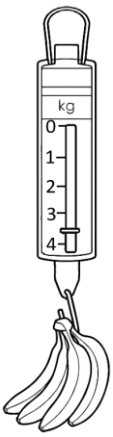
Reading: _____

b)



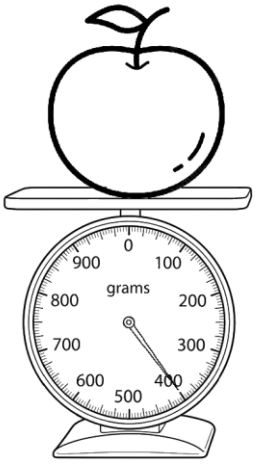
Reading: _____

c)



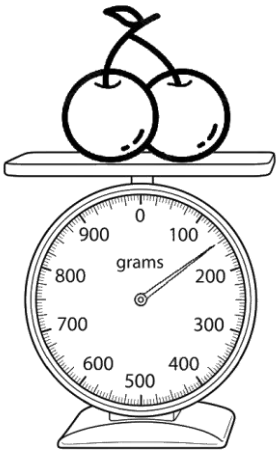
Reading: _____

d)



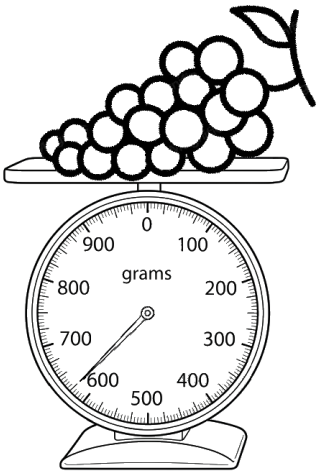
Reading: _____

e)



Reading: _____

f)



Reading: _____

3. Answer the questions

3.1) Elizabeth steps onto the scale and weighs **48 kg**. Next, she weighs her cat separately, and it weighs **4 kg**.
If she steps onto the scale while holding her cat, what combined weight will the scale display?

Answer: _____

3.2) Tom wants to find the weight of a chocolate slab using a balance scale.

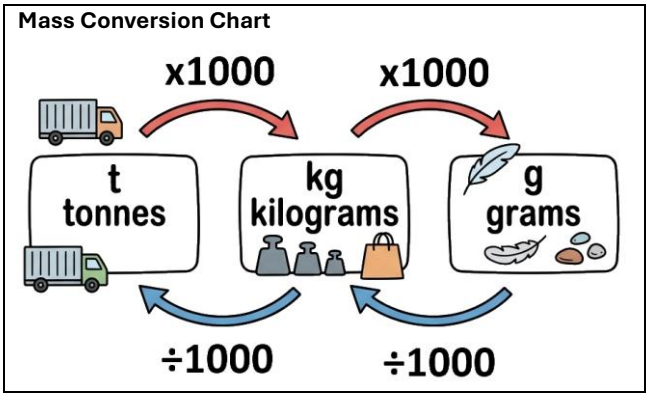
- First, he places the chocolate slab on one side of the scale, and one **50g weight** and three **20g weights** on the other side. The scale tips toward the weights because they are too heavy.
- Next, he replaces one of the **20g weights** with a **10g weight**, and the scale perfectly balances.

What is the weight of the chocolate slab?

Answer: _____

Conversion

Remember, “Horse to fly, multiply” and “Fly to horse, divide of course”



4. Convert the following units

a) 1,3 kg = _____ g

b) 4 500 g = _____ kg

c) 0,08 kg = _____ g

d) 2 t = _____ kg

e) 630 g = _____ kg

f) 3,75 t = _____ kg

g) 15 210 g = _____ kg

h) 8 400 kg = _____ t

i) 0,5 t = _____ g

j) 2 845 g = _____ kg

k) 0,45 kg = _____ g

l) 7 800 g = _____ kg

m) 5,2 t = _____ kg

n) 950 g = _____ kg

o) 12 500 kg = _____ t

p) 0,006 t = _____ kg

q) 3 050 g = _____ kg

r) 1,8 t = _____ g

s) 420 kg = _____ t

t) 0,075 kg = _____ g

Fraction questions

Method is to first convert the mass into smaller units first

Example: $3\ \frac{1}{2}kg$ = _____ g?

First 1kg = 1000g
For the **whole numbers**, there are 3 kg, so it's 3000g
For the **fraction**, what is a half of 1000g? It's 500g
The total is then $3000g + 500g\ =\ 3500g$

Note: if $\frac{1}{4}$ kg is 250g, then 3 quarters are ($\frac{3}{4}\ =\ 3 * 250g = 750g$) **OR** you can convert it to a decimal: $\frac{3}{4}kg = \frac{75}{100}kg = 0.75kg = 750g$

5. Find the unit conversions

a) $2\frac{1}{2}kg$ into grams

Whole part: _____ g

Fraction part: _____ g

Total: _____ g

b) $1\frac{1}{4}kg$ into grams

Whole part: _____ g

Fraction part: _____ g

Total: _____ g

c) $3\frac{1}{10}t$ into kilograms

Whole part: _____ kg

Fraction part: _____ kg

Total: _____ kg

d) $4\frac{3}{4}t$ into kilograms

Whole part: _____ kg

Fraction part: _____ kg

Total: _____ kg

Compare the following masses

6. Use <, > or = to compare the following. Remember we can only compare like units, so first convert them to the same unit.

a) 150 g _____ 0,1 kg

b) 2,5 kg _____ 2 500 g

c) 450 g _____ 0,5 kg

d) 3 t _____ 3 200 kg

e) 0,08 kg _____ 800 g

f) 1 500 kg _____ 1,5 t

g) 0,75 t _____ 750 kg

h) 6 200 g _____ 6,2 kg

i) 0,004 t _____ 40 kg

j) 850 g _____ 0,085 kg

Order the following

First convert all to the same units then compare

Example: Order the following in ascending order
$\frac{1}{2}$ kg; 3 000g; 2,4kg; 150g

Remember we can only compare like units so first convert to all the same.
500g; 3 000g; 2400g; 150g

Now we can order them, use the original
150g; $\frac{1}{2}$ kg; 2,4kg; 3 000g;

7. Order the following in ascending order

a) $\frac{1}{2}$ kg ; 1 200 g ; 3,5 kg ; 800 g

Order: _____

b) 2 500 g ; $\frac{1}{4}$ kg ; 1,8 kg ; 600 g

Order: _____

c) $\frac{3}{4}$ kg ; 0,9 kg ; 2 100 g ; 600 g

Order: _____

d) 5,2 kg ; 750 g ; $\frac{4}{5}$ kg ; 520 g

Order: _____

Word problems

8. Answer the following questions

8.1) What is the total mass of 2,3 kg, 500 g and 3 kg 450 g?

Answer: _____

8.2) What mass is 500 g less than 4,32 kg?

Answer: _____

8.3) How much lighter is 345 g than 1,1 kg?

Answer: _____

8.4) The heaviest mass allowed on a ride at the fairground is 55 kg. Diana's mass is 28,1 kg and Dave's mass is 27 000 g. Can they go on the ride together?

Answer: _____

8.5) The average mass of one peach is 175 g. What is the mass of nine peaches in kilograms?

Answer: _____

8.6) (a) The mass of eight apples is 1 200 g. What is the average mass of one apple in kilograms?

Answer: _____

(b) A 5 kg bag of apples costs R65. What is the cost of 500 grams of apples?
Hint: find the price of 1kg

Answer: _____

8.7) Which has a greater mass: twenty-eight 35 g packets of chips or three 400 g bags?

Answer: _____

8.8) Dried dog food comes in 5.2 kg bags. A dog eats 400 g each day. How long will the food last?
hint: convert into grams

Answer: _____

8.9) The masses of the items in Nombulelo's shopping are 450 g, 0,25 kg, 650 g and 1,3 kg. The doctor said that she must not carry more than 4 kg. What is the heaviest mass of potatoes she can add to her bag?

Answer: _____

Mass and Weight - Worksheet Memo / Answer Key

1. Estimations
1.1 Mathematics textbook: 800 g
1.2 Single chicken egg: 60 g
1.3 Full-grown domestic cat: 4,5 kg
1.4 Single paperclip: 1 g
1.5 Slab of chocolate: 100 g
1.6 Standard building brick: 3 kg
1.7 Standard bicycle: 12 kg
1.8 Standard loaf of bread: 700 g
1.9 Fully grown male lion: 190 kg
1.10 Farm tractor: 2 700 kg

2. Instrument Readings
(Readings based on typical dial/scale markings shown on Page 3)
a) 71,25 kg
b) 150 g
c) 3.5kg
d) 400 g
e) 150 g
f) 650 g

3. Word Problems (Page 3)
3.1) $48\text{ kg} + 4\text{ kg} = \mathbf{52}$ kg
3.2) Scale balances with $50\text{ g} + 20\text{ g} + 20\text{ g} + 10\text{ g} = \mathbf{100}$ g

4. Unit Conversions (Page 4)
a) $1{,}3\text{ kg} = \mathbf{1}\,300$ g
b) $4\,500\text{ g} = \mathbf{4{,}5}$ kg
c) $0{,}08\text{ kg} = \mathbf{80}$ g
d) $2\text{ t} = \mathbf{2}\,000$ kg
e) $630\text{ g} = \mathbf{0{,}63}$ kg
f) $3{,}75\text{ t} = \mathbf{3}\,750$ kg
g) $15\,210\text{ g} = \mathbf{15{,}21}$ kg
h) $8\,400\text{ kg} = \mathbf{8{,}4}$ t
i) $0{,}5\text{ t} = \mathbf{500}\,000$ g
j) $2\,845\text{ g} = \mathbf{2{,}845}$ kg
k) $0{,}45\text{ kg} = \mathbf{450}$ g
l) $7\,800\text{ g} = \mathbf{7{,}8}$ kg
m) $5{,}2\text{ t} = \mathbf{5}\,200$ kg
n) $950\text{ g} = \mathbf{0{,}95}$ kg
o) $12\,500\text{ kg} = \mathbf{12{,}5}$ t
p) $0{,}006\text{ t} = \mathbf{6}$ kg
q) $3\,050\text{ g} = \mathbf{3{,}05}$ kg
r) $1{,}8\text{ t} = \mathbf{1}\,800\,000$ g
s) $420\text{ kg} = \mathbf{0{,}42}$ t
t) $0{,}075\text{ kg} = \mathbf{75}$ g

5. Fraction Conversions (Page 4)
a) $2\frac{1}{2}$ kg into grams:
Whole part: 2 000 g
Fraction part: 500 g
Total: 2 500 g

b) $1\frac{1}{4}$ kg into grams:
Whole part: 1 000 g
Fraction part: 250 g
Total: 1 250 g

c) $3\frac{1}{10}$ t into kilograms:
Whole part: 3 000 kg
Fraction part: 100 kg
Total: 3 100 kg

d) $4\frac{3}{4}$ t into kilograms:
Whole part: 4 000 kg
Fraction part: 750 kg
Total: 4 750 kg

6. Comparing Masses (Page 5)
a) $150\text{ g} > 0{,}1\text{ kg}\,(100\text{ g})$
b) $2{,}5\text{ kg} = 2\,500\text{ g}$
c) $450\text{ g} < 0{,}5\text{ kg}\,(500\text{ g})$
d) $3\text{ t} < 3\,200\text{ kg}\,(3{,}2\text{ t})$
e) $0{,}08\text{ kg} < 800\text{ g}\,(80\text{ g} < 800\text{ g})$
f) $1\,500\text{ kg} = 1{,}5\text{ t}$
g) $0{,}75\text{ t} = 750\text{ kg}$
h) $6\,200\text{ g} = 6{,}2\text{ kg}$
i) $0{,}004\text{ t} < 40\text{ kg}\,(4\text{ kg} < 40\text{ kg})$
j) $850\text{ g} > 0{,}085\text{ kg}\,(85\text{ g})$

7. Ordering in Ascending Order (Page 5)
(Assuming missing fractions are restored as standard curriculum examples)
a) $\frac{1}{2}$ kg (500 g); 1 200 g; 3,5 kg (3 500 g); 800 g
Order: $\mathbf{\frac{1}{2}}$ kg; **800** g; **1** 200 g; **3,5** kg

b) 2 500 g; $\frac{1}{4}$ kg; 1,8 kg; 600 g
Order: $\mathbf{\frac{1}{4}}$ kg; **600** g; **1,8** kg; **2** 500 g

c) $\frac{3}{4}$ kg; 0,9 kg; 2 100 g; 600 g
Order: 600 g; $\mathbf{\frac{3}{4}}$ kg; **0,9** kg; **2** 100 g

d) 5,2 kg; 750 g; $\frac{4}{5}kg$; 520 g
Order: 520 g; **750** g; **800** g; **5,2** kg

8. Word Problems (Page 6)
8.1) $2{,}3\text{ kg} + 0{,}5\text{ kg} + 3{,}45\text{ kg} = \mathbf{6{,}25}$ kg *(or* 6 kg 250 g*)*
8.2) $4{,}32\text{ kg} - 0{,}5\text{ kg} = \mathbf{3{,}82}$ kg *(or* 3 820 g*)*
8.3) $1\,100\text{ g} - 345\text{ g} = \mathbf{755}$ g *(or* 0,755 kg*)*
8.4)$27\,000\text{ g} = 27\text{ kg}$
Total mass: $28{,}1\text{ kg} + 27\text{ kg} = 55{,}1\text{ kg}$
Answer: No, because 55,1 kg exceeds the 55 kg limit.

8.5) $175\text{ g} \times 9 = 1\,575\text{ g} = \mathbf{1{,}575}$ kg

8.6)
(a) $1\,200\text{ g} \div 8 = 150\text{ g} = \mathbf{0{,}15}$ kg
(b) $1kg = R13$. $R65 \div 2 =$ R**6,50**

8.7)
Chips: $28 \times 35\text{ g} = 980\text{ g}$
Bags: $3 \times 400\text{ g} = 1\,200\text{ g}$
Answer: Three 400 g bags have a greater mass.

8.8) 5.2 kg = 5 200 g. 5 200 g ÷ 400 g/day = **13** days
8.9) Current shopping: $0{,}45\text{ kg} + 0{,}25\text{ kg} + 0{,}65\text{ kg} + 1{,}3\text{ kg} = 2{,}65\text{ kg}$
Max remaining capacity: $4\text{ kg} - 2{,}65\text{ kg} = \mathbf{1{,}35}$ kg *(or* 1 350 g*)*